COMP-8117 · SPLASH PROPOSAL · DRAFT

Course materials every student can read

Turning lecture slides into files that blind and low-vision students can use with screen readers and braille displays, on the same day as everyone else.

The problem

Blind and low-vision students read course materials with screen readers (speech) and braille displays. Both only work if a file is properly structured. Most lecture PDFs and slides are not: the reading order is scrambled, diagrams are read as just “image”, math comes out as gibberish, and tables become a jumble of numbers.

The fix today is the accessibility office converting files by hand, which takes days. Sighted classmates have the material instantly.

> **Blind students fall behind, not because of ability, but because of file formats.**

What exists, and why it isn't enough

TODAY'S APPROACH	WHERE IT FAILS
Adobe's accessibility checker	Finds problems, doesn't fix diagrams or math
Auto-tagging tools	Work on plain text, break on slides, math and tables
Staff converting by hand	Accurate but slow, can't keep up
Asking instructors to fix it	Most don't know how or don't have time

Each tool solves one piece. Nobody has fixed the journey from the instructor's file to a student who can read it on time.

Our approach: investigate, then build

1. **Investigate (weeks 6–7).** Interview the accessibility office, blind and low-vision students and instructors. Run real course files through today's tools and record where they break. Measure how long students wait.
2. **Decide (week 7).** The evidence chooses the fix. Working focus: math- and diagram-heavy lecture slides, turned into accessible HTML that works with screen readers and braille displays.
3. **Build and test (weeks 8–10).** Fix reading order and structure, add AI-written diagram descriptions, convert math to MathML, and flag anything uncertain for a human to check.
4. **Prove it (weeks 9–10).** Real files, real students, before and after.

We don't build braille output. Existing tools (Liblouis, MathCAT) handle that once content is structured correctly. This is a deliberate, recorded decision.

How we'll know it worked

MEASURE	BEFORE	GOAL
Time until a student can use the file	Days	Same day
Accessibility checker score	Measured on real files	Large improvement
Screen-reader tasks completed (find a formula, explain a diagram, read a table)	Tested with real users	Clear improvement
Staff time per document	Measured	Much less
Accuracy of diagram descriptions	—	Checked by people, errors counted

Exact targets are fixed after we measure the baseline, before we build.

Team (each role essential)

ROLE	OWNER	OWNS
Research & users	[name]	Interviews, user testing, before/after measurement
Document processing	[name]	Layout, reading order, tables, HTML output
AI & math	[name]	Diagram descriptions, math conversion, accuracy checks
Platform & quality	Abhinav	Pipeline, deployment, CI, budget, rubrics, decisions log

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Meeting the defense requirements

DEFENSE POINT	HOW WE COVER IT
1. The problem is real	Accessibility office interviews, measured turnaround, real files that fail
2. The intervention fits	HTML and human review chosen because of our findings, recorded in the decisions log
3. The product works	Real course files converted and used by real screen-reader users
4. The method protects quality	Staged testing, accuracy checks, CI, budget tracking, adapting to evidence
5. The problem is reduced	Before/after numbers against targets set in advance
6. Real, scalable impact	Equal access; scales to other departments, universities, and the learning platform at upload
7. Evidence-based reasoning	Every claim tied to a measurement, interview or log entry

How it earns marks

GRADED COMPONENT	HOW WE EARN IT
Product quality (15%)	Original (fixes diagrams and math where tools fail), works on real files, measured impact
Methodology quality (15%)	Investigate first, decisions log, targets set in advance, budget and hours tracked
Reflection (10%, individual)	Each person owns a hard decision to reflect on
Competency floor	Learning Outcomes map tying each competency to evidence
Judgment	Our own rubric, compared with the teaching team's

Timeline

WHEN	WHAT
Oct 26–27	Email the accessibility office, ask instructors for files, first baseline scores
Oct 28	**Expert Review 2:** problem, early evidence, plan
Weeks 7–8	Interviews, analysis of where files break, decision, first pipeline
Weeks 9–10	Real-file conversion, user testing, before/after (ends Nov 25)
Week 11	Final analysis, reports
Week 12	Defense: live demo plus evidence

Risks

RISK	PLAN
Accessibility office slow to reply	Email now; instructors and students can still provide files and testing
Few blind testers	3–5 users is enough; accessibility staff can also test
AI describes diagrams wrongly	Measure accuracy, flag uncertain items for human review
Privacy of course files	Use only files we have permission to use

Budget: about $50–100 (AI usage and hosting), plus team hours tracked weekly.